Break Into Aerospace

Deutsches Zentrum für Luft- und Raumfahrt (DLR)

Engineering & Research Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

> *DLR is one of the most structurally complex research organisations in Europe — simultaneously a Helmholtz research centre, Germany's space agency, and one of Europe's largest project management agencies — with an employment model that differs from every other organisation in this series in ways candidates consistently underestimate. This guide consolidates the preparation tasks, signal areas, and strategic context needed to compete effectively across all relevant roles at all three of DLR's institutional functions.*

Critical Structural Note · DLR's Three Institutional Hats

DLR wears three distinct institutional hats simultaneously, and confusing them is the most common mistake candidates make. Identify your target hat before preparing.

Function	Description
Hat 1 — Forschungszentrum	Germany's national centre for aerospace, energy, and transportation research. Approx. 10,000 employees including 3,650 scientists at 35 locations; research budget of €1 billion, with ∼49% from competitively allocated third-party funds (*Drittmittel*). Most engineering and scientific staff work here.
Hat 2 — Deutsche Raumfahrtagentur	The German Space Agency at DLR coordinates German space activities at national and European levels, manages Germany's contributions to ESA and EUMETSAT, and administers ∼€860 million in German ESA funds. Staff are programme managers and policy specialists — closer to UKSA than to JPL.
Hat 3 — Projektträger	DLR as project management agency coordinates and manages €1.279 billion in research on behalf of German federal ministries (BMBF, BMWi/BMWK, and others). This function administers competitive grant programmes across multiple fields. Its professional culture differs entirely from the research function.

Part I · What DLR Actually Looks For

By Seniority Level

- **Student / Werkstudent / Intern** — The intern interview is typically a phone call covering the CV and subjects, followed by a video meeting with a brief technical element and a short candidate presentation. The key filter is whether your academic specialisation matches the institute's current research needs and whether you demonstrate genuine intellectual curiosity about the *specific* research being done. Generic enthusiasm for aerospace or space is insufficient. Speculative applications directly to institutes and research groups are explicitly encouraged; apply at least one semester in advance.

- **Wissenschaftliche Mitarbeiter / Postdoc / Research Engineer (0–6 years)** — Master's degree or Diplom is the baseline; PhD is preferred or required for senior scientific positions. DLR seeks scientists who have their own independent thinking and are ready to take on responsibility. Applications are institute-specific: you apply to a specific advertised position or make a speculative application to a specific research group. Fixed-term contracts are the norm — in some cases permanent positions arrive only after 15 years, or contracts expire without extension. This reflects the Drittmittel funding structure of German public research.

- **Senior Wissenschaftler / Abteilungsleiter / Institutsleiter** — Demonstrated independent research leadership is the filter: an established publication record, success in Drittmittel acquisition, experience supervising doctoral students and postdoctoral researchers, and in some cases significant international scientific reputation. The route to permanent senior positions typically passes through a successful habilitation or equivalent seniority signal combined with demonstrated Drittmittel success. Senior positions in the German Space Agency function have a different profile, closer to senior civil service with policy and programme management expertise.

Hands-On Signal Areas

Domain	Core Competencies Expected
Research Portfolio	PhD in a domain directly relevant to the target institute; peer-reviewed publications, conference proceedings (IAC/AIAA/ICAS). Your publication record is effectively your technical interview — the interview itself is more CV-driven and behavioural.
Drittmittel Literacy	Understanding of competitive third-party funding (ESA, Horizon Europe, BMBF, BMWi); for mid and senior positions, evidence of contributing to or leading successful competitive funding applications is a significant positive signal.
Language Competency	English is widely used in international programmes; professional German is a daily practical requirement for roles involving German industry partners, government committees, or domestic administration. The specific requirement varies by institute.
Infrastructure Fluency	Familiarity with large-scale research facilities (wind tunnels, propulsion test stands, flight simulators) and understanding the operational constraints they impose on experimental design; researchers who have used equivalent facilities are more immediately productive.
Systems Thinking	Ability to engage across discipline boundaries — connecting aerodynamics research to aircraft design, propulsion research to climate policy — aligned with DLR's multi-domain programme portfolio and international partnerships (ESA, NASA).

How DLR's Interview Format Differs from Other Organisations

- **Conversational and CV-driven, not structured-competency.** Unlike organisations with formal STAR rubrics or structured competency scoring, DLR's process is relatively informal and institute-specific. Technical depth is probed through discussion of your own published work, not textbook questions or whiteboard exercises.

- **Each application is to a specific institute and often a specific research group.** DLR has over 50 research institutes and facilities. The Institute of Aerodynamics in Göttingen is a different scientific community from the Institute of Space Systems in Bremen or the Department of Planetary Research in Berlin. Applying to "DLR centrally" is not how DLR hiring works.

- **The Motivationsschreiben must be specific to the institute and research group.** A generic motivation letter that could apply to any DLR institute is a pre-screening loss. The expected letter connects your research background specifically to the work of the target institute and, where possible, to the work of the specific group or project lead.
- **Salary is governed by TVöD** — the German public sector collective agreement. Pay scales (Entgeltgruppen) are transparent but significantly below comparable industry compensation in German aerospace. The trade-off is research freedom, infrastructure access, work-life balance, and the quality of the scientific problems.
- **There is no single "DLR interview."** There are 50 different interviews at 50 different institutes. An aerodynamics institute in Göttingen has different technical priorities and interview depth from a space systems institute in Bremen, an energy institute in Stuttgart, or the German Space Agency function in Bonn.

Part II · Top Preparation Tasks

Student / Werkstudent / Intern

4–5 h

1. Identify Your Specific Institute, Group, and Project — Apply Directly or Speculatively

Objective: Institute-specific targeting is the primary application strategy for DLR internships. Internships are available outside advertised positions through speculative applications directly to institutes. DLR explicitly encourages this route and recommends applying sooner rather than later. The most successful intern applications are those where a student has identified a specific research group whose publications align with their thesis topic and reached out directly.

Done Signal: You have identified at least three DLR institutes whose published research directly overlaps your academic specialisation, read 2–3 recent papers from each group, identified the group leader by name, and either applied to an advertised position or sent a targeted speculative application that references specific published work from that group and explains your specific academic connection to it.

3–4 h

2. Prepare a *Kurzpräsentation* of Your Thesis or Most Significant Project

Objective: DLR intern interviews include a brief presentation from the candidate, typically covering their research background and academic projects. The presentation must explain your work to an audience of researchers who are experts in an adjacent but not identical field. This is a research skill DLR values explicitly. Prepare versions in both German and English if applying to an international project.

Done Signal: You have a 5–7-minute presentation that opens with the research question in plain language, describes your methodology and why it was appropriate, presents one specific key finding, and closes with what the finding contributes to the broader field. You have delivered it to a colleague outside your specific subfield who confirmed the core concept was understood.

2–3 h

3. Write a Specific *Motivationsschreiben* Naming the Group and a Paper

Objective: DLR's application process requires a cover letter naming the job reference number and the specific institute. A letter that names the institute but could apply to any group within it is not specific enough. The expected letter names the specific research group or project, references one specific publication from that group, explains what your academic background adds to their current

2–3 h

research questions, and states a concrete contribution you could make during the internship.

Done Signal: You have a one-page *Motivationsschreiben* in German (with English translation if required) that names the specific institute and research group, references one specific publication and explains its contribution to your academic understanding, connects your thesis or project to a specific open research question the group is pursuing, and states one concrete method, dataset, or skill you bring that would advance their current work.

1–2 h

4. Understand DLR's Three-Function Structure

Objective: Candidates who apply to DLR thinking it is an aerospace company, a space operations agency, or a government ministry will consistently misframe their motivation and qualifications. The research function, space agency function, and Projektträger function each have genuinely different professional cultures, employment terms, and success criteria. Applying with the wrong mental model produces answers that misalign with what the specific function actually does.

Done Signal: You can explain the three DLR functions — research organisation, German Space Agency, and Projektträger — in under 3 minutes, state which function you are applying to, and explain why that specific function aligns with your professional interests and background. The explanation is specific enough that it could not be given unchanged to a candidate applying to a different DLR function.

1–2 h

5. Assess Your German Language Level Honestly for the Target Institute

Objective: DLR's international orientation means many research positions are accessible to non-German speakers, particularly in internationally partnered projects. However, the degree to which German is the working language varies significantly by institute. A position working on ESA-partnered satellite research will use English extensively; a position working with German aviation authorities will require professional German daily. The specific requirement is usually inferable from the project description and partner ecosystem.

Done Signal: You have assessed your German level (A1–C1/native), reviewed the target position's partner ecosystem and project description to estimate the actual German requirement, and made an honest judgement about whether your language level matches the role. If below B2 for a German-heavy role, you have either targeted a more international project or have a concrete improvement plan.

1–2 h

6. Understand TVöD Pay Grades and Make an Honest Career Trade-Off Assessment

Objective: DLR's salary is below comparable industry positions, and fixed-term contracts add employment uncertainty on top of the compensation differential. Candidates who have honestly evaluated the trade-off — research freedom, infrastructure access, publication opportunities, international collaboration, and work-life balance in exchange for lower compensation and fixed-term tenure — make more stable and productive DLR employees.

Done Signal: You have looked up the relevant TVöD Entgeltgruppe for your target role (E13 for Master's/Diplom, E14 for PhD, E15 for senior scientists), calculated the net monthly salary at your expected Stufe, compared it honestly to equivalent industry offers in the same region and field, and arrived at a genuine decision about whether the research-for-compensation trade-off aligns with your current priorities.

Wissenschaftliche Mitarbeiter / Postdoc / Research Engineer

All intern tasks still apply. The *Kurzpräsentation* must cover doctoral research or equivalent professional work. The *Motivationsschreiben* must engage with the specific group's current research questions at depth, not just name publications.

7. Build a Publication Record That Directly Demonstrates Domain Expertise

Ongoing + 2–3 h

Objective: DLR interviews are more behavioural than deeply technical during the interview itself — much of the technical weight rests on your CV and academic record. This means the work you have done before applying — the papers you have published, the conference presentations you have given, the datasets you have contributed — carries the technical credibility that other employers probe during the interview. Your publication record *is* your technical interview.

Done Signal: You can point to at least two peer-reviewed publications (or one first-author paper and one conference proceedings paper) that directly address questions in the target institute's domain. If your publications are in adjacent fields, you can articulate specifically how the methods or findings transfer to the target domain. You have a 2-minute explanation of your most cited or most impactful publication ready for any conversation that arises.

8. Study the Target Institute's Last Three Years of Publications In Depth

6–8 h

Objective: The group leader or institute director interviewing you has spent their career on the research questions their group works on. A candidate who has read the group's recent publications — not just skimmed abstracts but understood the methodology and findings of 3–4 papers — is having a peer conversation rather than a job application. The group leader will notice whether you have engaged with their actual work.

Done Signal: For the specific research group you are targeting, you have read 3–4 recent papers (within the last three years), can summarise the key contribution of each in one sentence, can identify the open question each paper leaves, and have prepared two specific technical questions that reference the actual work — not generic questions about the institute's research area. You can discuss these questions in German if the role requires it.

9. Prepare a Specific Research Proposal or Contribution Statement

3–4 h

Objective: DLR seeks scientists who have their own independent thinking and want to go their own way. A candidate who arrives with a prepared view of what research question they would pursue, why it is scientifically important, and how they would approach it methodologically is demonstrating the research independence DLR values. Saying "I'll work on whatever you need" demonstrates execution willingness but not research leadership — insufficient for a scientific career track.

Done Signal: You have a 3-minute prepared statement of a specific research question you would pursue at the target institute — the question is named, its scientific significance is explained in one sentence, the methodology is described, the specific DLR infrastructure or collaboration that makes the question workable is named, and one concrete expected output (a publication type, a dataset, a technology demonstrator) is stated. The question is genuinely novel and not a restatement of work the group is already publishing.

10. Understand DLR's Drittmittel Funding Model and Career Trajectory Implications

2–3 h

Objective: Approximately 49% of DLR's research budget comes from competitively allocated third-party funds. This means roughly half of DLR's research staff work on positions funded by competitive grants with finite project durations. Understanding which funding source your target position uses, what the grant's remaining duration is, and what your career options are when the grant ends is not paranoia — it is research career literacy.

Done Signal: You can explain what Drittmittel means in the German research context, name three common Drittmittel sources for DLR positions (ESA, BMBF, Horizon Europe, BMWi, industrial contracts), understand the Wissenschaftszeitvertragsgesetz (WissZeitVG) and how it limits total fixed-term contract duration, and have a realistic view of your next career step if the initial DLR position does not convert to permanent employment.

11. Prepare a Story About Independent Scientific Problem Identification and Pursuit

1–2 h

Objective: DLR is a research institution, not a product development organisation. Its scientific staff are expected to identify problems worth studying, propose methods to study them, and pursue results that advance the field. Mid-level candidates who can demonstrate they have done this — identified an interesting question, designed an investigation, obtained unexpected results, interpreted them, and published — distinguish themselves from technically capable execution resources.

Done Signal: You have a story about a research direction you personally identified and pursued — the question was not assigned to you by a supervisor, the investigation approach was your own design, you encountered an unexpected result and revised your understanding, and the work produced a specific output (a paper, a report, a dataset, a new method). The story names the question, your reasoning for its importance, and the outcome.

12. Study DLR's Major Ongoing Programmes Relevant to the Target Institute

3 h

Objective: DLR researchers work within a broader programme context — national space programme priorities, ESA programme participation, German aviation technology priorities (LuFo programme), and Helmholtz Association research themes. DLR has contributed to Columbus, TanDEM-X, Galileo, Mars Express, Rosetta, and Cassini-Huygens, among others. A researcher whose work connects to DLR's national programme obligations is more institutionally valuable than one working on a question with no programme connection.

Done Signal: For the research area of your target institute, you can name one or two active DLR programmes or ESA missions the institute contributes to, explain what technical or scientific question the institute's contribution addresses, and describe one specific research challenge in that programme that your background is directly relevant to. You can do this in the language of the role.

13. Understand the Forschungszentrum vs. Deutsche Raumfahrtagentur Culture Distinction

2 h

Objective: The German Space Agency function (Bonn and Cologne) manages Germany's ESA contributions and national space programme — it is a programme management and policy function whose staff interface with ESA programme boards, BMWi/BMWK, and German industry. The research institute function develops original science and technology. Both carry the DLR name but have genuinely different professional identities, promotion tracks, skills profiles, and daily working lives.

Done Signal: You can describe the German Space Agency function in plain terms (what it manages, who its customers are, what its staff do daily), contrast it with the research institute function, and ar-

2 h

ticulate clearly which culture you are entering with your specific application — and why that culture fits your professional identity and career goals.

Senior Wissenschaftler / Abteilungsleiter / Gruppenleiter

All mid-level tasks still apply. The publication record must be substantive and show a consistent research agenda, not scattered contributions. The Drittmittel understanding must include personal experience applying for or contributing to competitive funding proposals.

4–6 h

14. Walk Through Your Research Programme — A Coherent Narrative Over Years

Objective: Senior DLR scientific positions require candidates who have an established research programme — a coherent intellectual agenda that their publications have advanced over several years, with a clear direction for where this programme leads next. The interview at this level probes whether your research has internal logic and cumulative momentum, or whether you have produced technically competent but disconnected outputs. The group leader interviewing you is evaluating whether you can build and lead a research group advancing a coherent scientific agenda.

Done Signal: You can describe your research programme in 5 minutes — the central question that motivates it, the sequence of papers that have advanced it, your most significant methodological or empirical contribution, the current open frontier, and the next two projects that would advance it. The narrative has internal coherence: each paper connects to the next, rather than appearing as a collection of separate projects.

2–3 h

15. Prepare Specific Drittmittel Acquisition Examples and a Next Proposal at DLR

Objective: Senior researchers who can attract Drittmittel — from ESA, Horizon Europe, BMBF, or industry — are directly funding their own teams and enabling the research programme they want to pursue. DLR group leaders who cannot attract external funding are dependent on institutional allocation and cannot grow their groups. This makes Drittmittel acquisition capability the senior research career metric above almost all others at DLR.

Done Signal: You have a prepared story about a competitive research proposal you led or co-led — the funding source, the call you responded to, your specific contribution to the proposal, the reviewers' feedback, and whether it was funded. You can also articulate one specific Drittmittel opportunity you would pursue at DLR and why your scientific background positions you to win it.

1–2 h

16. Prepare Specific Examples of Doctoral Student Supervision and Outcomes

Objective: DLR group leaders and senior scientists are expected to supervise doctoral candidates and postdoctoral researchers, contributing to Germany's scientific workforce development as part of the Helmholtz Association's educational mission. The probe is not whether you have supervised students — it is whether your supervision produced independent researchers who now contribute to the field, and whether you can articulate a specific approach to developing research independence rather than managing task execution.

Done Signal: You have a story about supervising a doctoral candidate — their research question, your approach to developing their independence, a specific moment when you shifted from directing to advising, the outcome of their thesis, and their subsequent career step. The story demonstrates that your supervision produced an independent scientist, not a project executor.

17. Prepare a Technically Grounded Position on an Open Research Frontier in Your Domain

4–5 h

Objective: Senior DLR scientific hiring is in part a bet on a research direction — the institute is investing institutional resources in someone who will shape a research agenda for years. Candidates who can articulate a specific open scientific question, explain why current methods are insufficient, describe a new approach that could work, and connect DLR's specific infrastructure or partnerships to why DLR is the right place to pursue it are making the strongest possible case for their hire. This is the intellectual pitch that justifies the institutional investment.

Done Signal: You have a prepared 5-minute discussion of one open frontier in your research domain — named specifically, current knowledge characterised honestly (what is known, what is not), a new approach that could advance the question described, and DLR's specific contribution (infrastructure, existing expertise, international partnerships, or programme relevance) that makes the pursuit at DLR specifically compelling. The discussion is grounded enough that a domain expert could challenge specific claims.

18. Build Familiarity with DLR's Research Infrastructure and Articulate Specific Planned Use

3 h

Objective: One of DLR's most significant competitive advantages over universities is its large-scale research infrastructure — the European Transonic Windtunnel (ETW, Cologne), propulsion test stands at Lampoldshausen, flight facilities at Oberpfaffenhofen and Braunschweig, the Columbus Control Centre, and Europe's largest civilian research fleet. Senior researchers who have specific plans to use this infrastructure — not just generic awareness that it exists — are demonstrating they are applying to DLR for what it specifically enables.

Done Signal: For the target institute, you can name the specific research infrastructure most relevant to your work, explain what experiment it would enable that you could not conduct at a university, and describe one specific research question you would address using that infrastructure. The description is specific enough that an infrastructure operator could confirm whether your planned use is feasible.

All Levels

19. "Warum DLR — und warum dieses Institut?" — Prepare a Specific, Non-Generic Answer

1–2 h

Objective: This question is asked in every DLR interview at every level. The expected answer is not "because DLR is one of Europe's leading research organisations" — it is a specific answer that names the institute, names the research group or project, explains one specific scientific question that this group works on that cannot be adequately pursued elsewhere, and connects your academic background to that question. The answer must be specific enough that it could not be given unchanged to a different DLR institute.

Done Signal: You have a 90-second answer in German (or English for international applications) that names the specific DLR institute and research group, explains one specific research question or infrastructure capability that makes this group unique, connects your academic work to that specific question or capability, and states clearly what you would contribute — and what you would learn — that is only possible at this specific DLR institute.

20. Verify Work Authorisation in Germany and Assess the WissZeitVG Fixed-Term Landscape

2 h

Objective: International candidates must have the right to work in Germany (EU citizenship, Blue Card, job seeker visa, existing work permit, or employer sponsorship). The Wissenschaftszeitvertragsgesetz (WissZeitVG) limits the total duration of fixed-term scientific employment contracts in

2 h

Germany — typically six years pre-PhD and six years post-PhD. Candidates who have already used significant fixed-term contract time at German universities or other Helmholtz centres have less remaining contractual flexibility. Understanding this before applying prevents uncomfortable conversations after an offer is made.

Done Signal: You know your work authorisation status in Germany, understand WissZeitVG limitations and how much of your allowed fixed-term contract time has been used, can answer standard DLR HR questions about contract duration and career plans, and have a realistic expectation of what a 2–3-year DLR research position means for your subsequent career step — whether in German academia, European industry, or elsewhere.

Summary Reference Table

#	Task Description	Target Level	Effort
1	Identify specific institute/group — apply directly or speculatively	All	4–5 h
2	*Kurzpräsentation* (5–7 min, German and English)	All	3–4 h
3	*Motivationsschreiben* naming institute, group, and paper	All	2–3 h
4	Three-function structure: research, space agency, Projektträger	All	1–2 h
5	German language level — honest assessment for target institute	All	1–2 h
6	TVöD pay grades — honest career trade-off assessment	All	1–2 h
7	Publication record — peer-reviewed papers in target domain	Mid+	Ongoing + 2–3 h
8	Target group's recent publications — 3–4 papers in depth	Mid+	6–8 h
9	Research proposal or contribution statement — specific and novel	Mid+	3–4 h
10	Drittmittel funding model and career trajectory implications	Mid+	2–3 h
11	Independent problem identification and research pursuit story	Mid+	1–2 h
12	DLR national programme context for target institute	Mid+	3 h
13	Forschungszentrum vs. Deutsche Raumfahrtagentur culture distinction	Mid+	2 h
14	Research programme walk-through — coherent narrative over years	Senior	4–6 h
15	Drittmittel acquisition examples and next proposal at DLR	Senior	2–3 h
16	Doctoral student supervision — independence-building story	Senior	1–2 h
17	Open research frontier — DLR-specific contribution position	Senior	4–5 h
18	DLR research infrastructure — specific planned use	Senior	3 h
19	"Warum DLR — warum dieses Institut?" — specific, not generic	All	1–2 h
20	Work authorisation, WissZeitVG, and fixed-term career planning	All	2 h

The Structural Reality

DLR was founded on a heritage stretching back to Ludwig Prandtl's aerodynamics laboratory in Göttingen in 1907 — the same institution where boundary layer theory, the foundation of all modern aerodynamics, was developed. Over a century of accumulated research infrastructure, scientific method, and institutional knowledge has been built on that foundation, and DLR now operates it simultaneously on behalf of the German federal government, ESA, and the broader international scientific community.

The engineers and scientists who succeed at DLR are those who find genuine fulfilment in a specific mode of professional contribution: where the output is knowledge, the metric is publication, citation, and grant success, and the time horizon is measured in research careers rather than product cycles. Those who struggle are those who discover that public research employment moves at a different pace from industry, compensates at a different level, and offers a different kind of security — and who have not made that trade-off honestly before arriving.

DLR's combination of Helmholtz-scale infrastructure, German Space Agency mandate, and Projektträger function makes it one of the few places in Europe where a researcher can move, across a career, from basic aerodynamic research to space mission support to national space policy implementation — all within a single institutional identity. That breadth is rare. **Demonstrate that you understand it, and that you are choosing it deliberately.**

BREAK INTO AEROSPACE